Joint Fragment Dissemination and Edge Fusion for Fast Target Detection in UAV-Assisted Urban IoT

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***Abstract*—Urban edge-camera and IoT sensing networks increasingly support time-critical monitoring, where rapid target detection is essential in emergency scenarios. In these environments, Unmanned Aerial Vehicles (UAVs) serve as mobile data ferries to disseminate high-value recognition payloads to distributed ground nodes. However, large payloads are severely constrained by short air-to-ground (A2G) contact windows and packet-level impairments such as path loss, shadowing, and stochastic packet errors (PER).**

This paper addresses the problem of minimizing mission-level time-to-detection under realistic PHY-layer constraints. We propose a coordinated dissemination framework that partitions recognition payloads into small fragments of comparable information content. A UAV continuously broadcasts these fragments (rather than the full payload) along a coverage-aware trajectory, while ground nodes cooperatively exchange and fuse missing evidence through localized collaboration. Unlike conventional UAV-only delivery, our design explicitly coordinates fragment broadcasting with edge cooperation under packet-level feasibility.

Simulation results show that the proposed framework significantly reduces time-to-detection compared to monolithic transmission and UAV-only baselines, particularly under stringent A2G conditions with short contact windows and high packet error rates. The results highlight the value of jointly optimizing fine-grained dissemination, edge cooperation, and PHY-aware communication in UAV-assisted urban sensing systems.

***Index Terms*—UAV-assisted dissemination, edge cooperation, fragment-based recognition, low-latency detection**

I. Introduction

UAV-assisted IoT systems have emerged as a practical solution in many real-world domains. UAVs serve as flexible mobile platforms that support environmental monitoring, industrial data collection, smart agriculture, and urban intelligent systems. Thanks to their high-altitude flight and wide coverage capability, UAVs enable connectivity with spatially distributed IoT sensor nodes over large areas without relying entirely on fixed infrastructure.

The target detection task is a fundamental problem in UAV-assisted IoT systems, particularly in time-sensitive monitoring scenarios. We consider a sensing region populated by a set of ground nodes $\mathcal{N}$, where a subset of nodes $\mathcal{P} \subseteq \mathcal{N}$ is identified as candidate nodes associated with a potential event of interest. Among them, a single unknown node n^* is the designated detection node that must accumulate sufficient evidence to trigger detection.

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The UAV disseminates recognition data to ground nodes during flight, while nodes can further exchange received information through local cooperation. Each node incrementally builds a confidence score as it collects useful data. The objective is to enable the unknown node n^* to reach a detection threshold as early as possible, under limited air-to-ground contact time and unreliable wireless links.

Traditional methods typically adopt a straightforward approach: the UAV flies close to each node and transmits the entire payload in a single session. However, this strategy requires the UAV to fly slowly or hover for extended periods to guarantee complete data delivery, which significantly increases mission duration and delays detection. Moreover, realistic physical-layer (PHY) constraints must be considered, such as short air-to-ground contact windows, path loss, shadowing, and stochastic packet errors (PER). These factors not only increase the UAV's energy consumption but also substantially reduce overall system efficiency. Therefore, a faster and more effective approach is needed.

We propose a coordinated approach in which the UAV broadcasts compact fragments continuously along a coverage-aware path, while ground nodes perform selective edge cooperation to share missing fragments. This design helps the designated detection node accumulate evidence earlier without requiring full payload delivery to every node.

The core ideas include payload fragmentation, cell-based localized cooperation, selective fragment broadcasting along an optimized trajectory, and edge exchange of missing fragments. These components are integrated in one coordination framework to reduce detection time while keeping UAV flight cost low.

To the best of our knowledge, no prior work has jointly considered recognition payload fragmentation, coverage-aware UAV trajectory planning, and edge cooperation under realistic packet-level PHY constraints.

The remainder of the paper is organized as follows. Section II reviews related work and positions our contribution. Section III describes the system model. Section IV formulates the mission-time minimization problem. Section V presents the proposed coordination protocol in detail. Section VI evaluates performance through extensive simulations. Section VII discusses limitations and future extensions, and Section VIII concludes the paper.

II. Related Work

Recent studies have widely employed UAVs as mobile platforms (data ferries) or aerial base stations to support data collection and dissemination in IoT networks [1], [2]. These works primarily focus on optimizing UAV trajectories to improve coverage, Age-of-Information (AoI), or detection latency [3], [4]. However, most approaches assume monolithic payloads and idealized channel models (binary in-range), overlooking realistic physical-layer constraints such as short contact windows and stochastic packet error rates (PER) in urban air-to-ground (A2G) environments. Consequently, these solutions are often unsuitable for transmitting high-quality recognition payloads under harsh channel conditions.

In a parallel line of research, distributed edge inference and cooperative detection aim to enhance detection accuracy through local information exchange and consensus mechanisms [5], [6]. Recent works have also explored early-exit techniques and confidence-threshold mechanisms to reduce decision latency. While these approaches are effective in environments with stable infrastructure, they typically assume negligible communication overhead and continuous connectivity. In the context of UAV flyovers with extremely short contact durations, these assumptions no longer hold, rendering edge cooperation inefficient or infeasible without proper coordination with the UAV dissemination process.

A third research direction investigates progressive or fragmented data delivery, a concept long studied for bandwidth-limited channels. Recent studies have demonstrated that even when images or feature maps are fragmented, each fragment retains meaningful recognition value and can be flexibly fused using confidence accumulation, Bayesian fusion, or log-odds methods [7], [8]. In edge-AI contexts, feature-map partitioning further enables early detection without requiring the full payload [9]–[11]. Nevertheless, these techniques have largely been developed independently of UAV mobility and have not been jointly optimized with realistic A2G channel constraints at the packet level.

In summary, the existing literature typically addresses the three pillars independently: UAV trajectory optimization focuses on mobility, edge cooperation emphasizes distributed inference, and fragmented delivery concentrates on data structure.

Building upon these foundations, this paper presents a coordinated framework that integrates the three key elements—recognition payload fragmentation, coverage-aware UAV trajectory planning, and edge cooperation—under realistic packet-level PHY constraints (short contact windows and stochastic PER). Simulation results demonstrate that the proposed framework significantly reduces mission-level time-to-detection, particularly under stringent A2G channel conditions.

III. System Model and Problem Formulation

A. System Model

Network Model. We consider an urban IoT system with a base station (BS), a UAV, and a set of ground nodes $\mathcal{N}$ distributed over a monitoring region and organized into cells for localized cooperation.

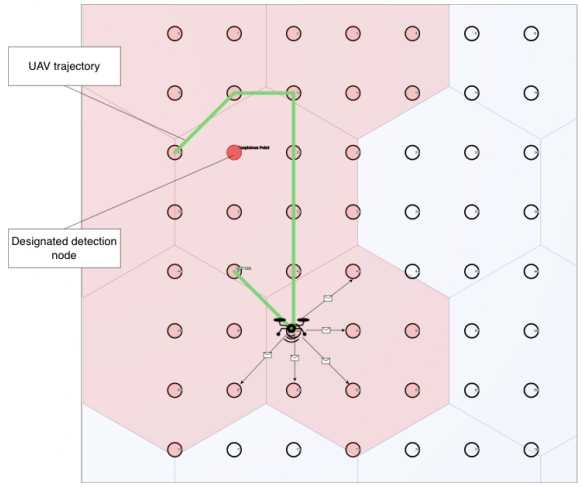

Fig. 1. System overview of UAV-assisted fragment dissemination and edge cooperation.

A subset $\mathcal{P} \subseteq \mathcal{N}$ represents candidate nodes associated with a potential event, among which an unknown node n^* must accumulate sufficient evidence for detection. The mission is completed when n^* reaches a predefined confidence threshold.

Each node can receive fragments from the UAV, exchange information with neighbors, and maintain a local confidence score.

Fig. 1 shows the UAV flight trajectory and fragment dissemination process. The UAV continuously broadcasts fragments along its path, while the highlighted red regions denote prioritized coverage zones where ground nodes receive and later cooperatively exchange information.

UAV Mobility and Coverage Model. The UAV flies at a fixed altitude h and follows a continuous waypoint sequence $\Pi = \{c_1, c_2, \ldots, c_M\}$ at constant speed v without hovering. Fragments are broadcast *continuously* along the entire trajectory while the UAV is in motion. Coverage at any time t is determined by the instantaneous position of the UAV and a broadcast radius R_b derived from the A2G link budget.

Let $\mathrm{CS}(t) \subseteq \mathcal{P}$ denote the set of candidate nodes covered by the UAV at time t. The trajectory is planned over a candidate set of locations (e.g., node positions or cell centroids) to maximize coverage efficiency while respecting mobility constraints.

The instantaneous Euclidean distance between the UAV and node n at time t is

$$d_n(t) = \sqrt{(x_{\mathrm{uav}}(t) - x_n)^2 + (y_{\mathrm{uav}}(t) - y_n)^2 + h^2}. \quad (1)$$

Data Dissemination and Cooperation Model. A recognition payload is partitioned into K fragments. The UAV continuously broadcasts these fragments during flight. Let $\mathcal{F}_n(t)$ denote the set of fragments successfully received by node n up to time t.

Ground nodes engage in cooperative exchange (both intra-cell and inter-cell via designated gateways) to share received

fragments and accelerate confidence accumulation at the designated detection node. The detailed triggering conditions and exchange protocols are specified in the coordination algorithms presented in Section V.

Fragment and Detection Model. Each fragment i provides an independent evidence contribution $p_i \in (0, 1)$. The confidence at node n accumulates over received fragments as

$$C_n(t) = 1 - \prod_{i \in \mathcal{F}_n(t)} (1 - p_i), \tag{2}$$

which captures the union of independent evidence.

In practice, this accumulation can be implemented in a log-odds form for numerical stability and efficient incremental updates, where each fragment contributes additively to the confidence score.

Communication and PHY Model. The UAV–node link follows a standard air-to-ground (A2G) channel model that accounts for distance-dependent path loss, log-normal shadowing, and small-scale fading [12]. The received power at node n at time t is

$$P_{\text{rx}}(t) = P_{\text{tx}} - \text{PL}(d_n(t)) - X_\sigma(t) - X_f(t), \tag{3}$$

where P_{tx} is the transmit power, $\text{PL}(\cdot)$ is the path-loss function, $X_\sigma(t)$ is log-normal shadowing, and $X_f(t)$ is small-scale fading.

The instantaneous signal-to-noise ratio (SNR) determines the bit error rate (BER). This BER is mapped to packet error rate (PER) via

$$\text{PER} = 1 - (1 - \text{BER})^{8L}, \tag{4}$$

where L is the packet size in bytes [12]. A packet is successfully received only if (i) $P_{\text{rx}}(t) \geq P_{\text{sens}}$ (receiver sensitivity threshold) and (ii) the condition holds throughout the entire packet transmission duration.

Timing and Contact Model. Let $t_{\text{tx}} = 8L/R$ denote the transmission time of one packet, where R is the data rate. Because the UAV broadcasts continuously while moving, a *contact window* between the UAV and node n is defined as the maximal continuous time interval during which $P_{\text{rx}}(t) \geq P_{\text{sens}}$. All fragment deliveries and cooperative exchanges must complete within these valid contact windows. Scheduling decisions therefore incorporate both instantaneous channel feasibility and per-node buffer constraints.

B. Problem Formulation

Let T_{detect} be the earliest time at which the designated detection node n^* reaches the alert threshold:

$$T_{\text{detect}} = \inf\{t \mid C_{n^*}(t) \geq \tau_{\text{alert}}\}. \tag{5}$$

The objective is to minimize the mission-level detection time:

$$\min_{\Pi, \mathcal{S}} \quad T_{\text{detect}}, \tag{6}$$

where Π is the UAV waypoint sequence and $\mathcal{S}$ is the joint scheduling policy for fragment broadcasting and cooperative exchange. The optimization is subject to UAV mobility constraints (speed and trajectory continuity), communication constraints (contact windows, PER, and receiver sensitivity), intra- and inter-cell cooperation limits, and per-node buffer/scheduling constraints.

IV. Proposed Approach

Overview. The proposed coordinated dissemination framework integrates three tightly coupled modules that operate in a closed loop: (i) coverage-aware UAV trajectory planning, (ii) continuous fragment broadcasting, and (iii) distributed edge cooperation. The base station (BS) first constructs a globally near-optimal UAV path using a greedy maximum-coverage (GMC) heuristic. While the UAV flies along this path, it continuously broadcasts recognition fragments. Ground nodes update their local confidence scores upon successful reception and trigger cooperative exchanges when necessary to accelerate evidence accumulation at the designated detection node n^*.

The framework takes as input the network topology, candidate-node set $\mathcal{P}$, threshold parameters τ_{coop} and τ_{alert}, and PHY settings. It outputs the UAV waypoint sequence Π, the transmission schedule $\mathcal{S}$, and the detection time T_{detect}. A heuristic design is adopted because the joint optimization of , scheduling, and random packet outcomes is computationally intractable for real-time missions.

Algorithm 1 summarizes the end-to-end coordination process, including cell initialization, trajectory planning, fragment dissemination, and confidence-driven cooperation. It serves as a high-level abstraction of the proposed system operation.

Cell Formation and Local Coordination Structure.

To support efficient localized cooperation under limited communication range, ground nodes are organized into geographic cells. This cell structure reduces coordination overhead, limits contention, and enables scalable fragment exchange within bounded neighborhoods.

Fig. 2 illustrates the cell partitioning, leader election outcome, and gateway-to-gateway inter-cell forwarding links used by the coordination protocol.

The cell formation and leader election procedure is summarized in Algorithm 2. Nodes are grouped based on spatial proximity, and a lightweight distributed selection mechanism is used to assign coordination roles within each cell.

UAV Trajectory Planning (GMC). To achieve efficient spatial coverage with minimal flight distance, we employ a greedy maximum-coverage (GMC) heuristic. Candidate waypoints are selected from node positions and cell centroids. The algorithm iteratively chooses the waypoint that provides the highest coverage gain per unit travel cost while considering cell-level coverage (if a waypoint covers more than a threshold fraction β of a cell, the entire cell is marked as covered). The resulting trajectory Π balances coverage efficiency and travel cost.

The pseudocode of the GMC planner is given in Algorithm 3.

Fragment Broadcasting and Confidence Update. Along the planned trajectory Π, the UAV continuously broadcasts the K fragments in a cyclic manner without hovering. Each ground

Algorithm 1 Coordinated Detection Workflow (Simplified)

Require: $\mathcal{G}, \mathcal{P}, \mathcal{D}$, PHY params, n^*
Ensure: $\Pi, \mathcal{S}, T_{\text{detect}}$

Initialize cells $\{\mathcal{C}_i\}$, leaders, and coverage map
Partition $\mathcal{D}$ into K fragments
Compute UAV path Π via GMC
Initialize $C_n \leftarrow 0, \forall n$
for each waypoint $w \in \Pi$ **do**
  UAV broadcasts fragments
  for each node n receiving new fragments **do**
    Update C_n
    if $C_n \geq \tau_{\text{coop}}$ **then**
      Trigger cooperative exchange
    end if
    if $n = n^*$ and $C_n \geq \tau_{\text{alert}}$ **then**
      return T_{detect}
    end if
  end for
end for

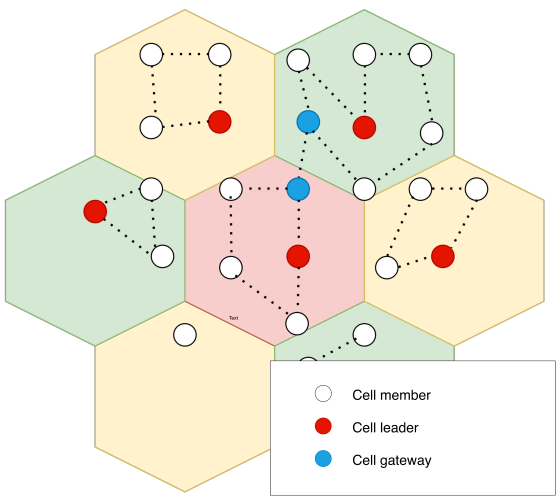

Fig. 2. Cell partitioning, leader election, and gateway-based inter-cell forwarding structure.

node n maintains a local set of received fragments $\mathcal{F}_n(t)$ and updates its confidence score $C_n(t)$ incrementally whenever a new fragment is successfully received (using either the union-probability or log-odds form presented in Section III).

Cooperative Edge Protocol. To compensate for packet losses caused by short contact windows, nodes engage in both intra-cell and inter-cell cooperation. Cooperation is triggered by three events: (i) local confidence exceeding τ_{coop}, (ii) the end of a UAV contact window, or (iii) receipt of a manifest revealing missing fragments. Upon triggering, a node broadcasts a compact manifest and exchanges missing fragments with peers. Cell gateways handle inter-cell forwarding when necessary, using timeout and jitter mechanisms to avoid collisions.

The complete cooperative exchange logic is detailed in Algorithm 4.

Algorithm 2 Cell Formation and Leader Election

Require: Node set $\mathcal{N}$ with positions
Ensure: Cell structure $\{\mathcal{C}_i\}$ with leaders

Partition nodes into cells based on geographic proximity
for each cell $\mathcal{C}_i$ **do**
  Nodes discover neighbors within communication range
  Compute fitness score for leader selection (e.g., proximity to cell centroid)
  Elect node with highest score as cell leader
  Designate gateway nodes for inter-cell communication
end for

Algorithm 3 Coverage-Aware UAV Path Planning (GMC)

Require: Candidate set $\mathcal{C}$, candidate-node set $\mathcal{P}$
Ensure: Ordered waypoint list Π

Initialize $\Pi \leftarrow \emptyset$, $\mathcal{P}_{\text{covered}} \leftarrow \emptyset$
while $\mathcal{P}_{\text{covered}} \neq \mathcal{P}$ **do**
  for each candidate $c \in \mathcal{C}$ **do**
    Compute coverage gain $G(c)$ over uncovered nodes
    if c covers $> \beta$ fraction of a cell **then**
      Mark entire cell as covered
    end if
    Compute cost $L(c)$ (travel distance)
    Score $S(c) \leftarrow \frac{G(c)}{L(c)^{\alpha}+\epsilon}$
  end for
  Select $c^* = \arg\max S(c)$
  Append c^* to Π
  Update $\mathcal{P}_{\text{covered}}$
end while

V. Performance Evaluation

A. Simulation Setup

We evaluate the proposed framework on grid-based urban IoT deployments with $N \in \{100, 400, 900, 1225\}$ nodes. Two UAV policies are compared: nearest-neighbor delivery (NN) and the proposed GMC-based dissemination with and without cooperation. All quantitative defaults are listed in Table I. For statistical stability, each configuration is repeated for 100 independent seeds; reported metrics are aggregated over completed runs.

We implement all experiments in the ns-3 discrete-event simulator, which enables detailed modeling of packet-level transmission, channel impairments, and A2G communication dynamics.

B. Metrics

To evaluate the effectiveness of the proposed fragment-based cooperative detection and GMC planning, we employ the following metrics:

- **Detection Time** (T_{detect}): The primary metric, defined as the time interval from UAV flight initialization until

Algorithm 4 Cooperative Exchange (Intra- and Inter-cell)

Require: Node n, fragment set $\mathcal{F}_n$, thresholds $\tau_{\text{coop}}, \tau_{\text{alert}}$, cell gateway (CGW), neighbor-cell gateway (NGW)

Ensure: Updated fragment sets and confidence

Trigger := ($C_n \geq \tau_{\text{coop}}$) OR (UAV contact ended) OR (peer manifest reveals missing fragments)
if Trigger **then**
    Broadcast manifest; receive requests; exchange missing fragments with neighbors (with collision-aware scheduling)
    Update $\mathcal{F}_n$ and C_n
end if
if cell gateway (CGW) detects stagnation (no new manifests) AND $\exists n : C_n < \tau_{\text{coop}}$ **then**
    if neighbor-cell gateway (NGW) has missing fragments AND target nodes satisfy $C_n < \tau_{\text{alert}}$ (after timeout) **then**
        Request and forward fragments via gateway-to-gateway coordination
        Update fragment sets and confidences across cells
    end if
end if

TABLE I
SIMULATION PARAMETERS USED IN ALL EXPERIMENTS (NOTATION: L=LOS, M=MIXED, N=NLOS)

Parameter	Symbol	Value
Topology		
Network sizes	N	100, 400, 900, 1225
Grid spacing	s	20 m
Candidate-node fraction	ρ	30% of nodes
Cell radius	R_{cell}	80 m
UAV		
Altitude	h	20 m
Speed	v	20 m/s
PHY (CC2420 / IEEE 802.15.4)		
Data rate	R	250 kbps
Elevation thresholds	θ	40° (L), 20° (M)
Packet size	L	100 bytes
Fragments and confidence		
Fragment count	K	10
Evidence range per fragment	p_i	[0.1, 0.9]
Master confidence target	C_{master}	0.90
Thresholds		
Cell coverage threshold	β	0.30
Cooperation threshold	τ_{coop}	0.30
Alert threshold	τ_{alert}	0.75
Cooperation stagger	—	20 ms/hop

the designated detection node reaches the alert threshold τ_{alert}.

- **UAV Path Length**: The cumulative distance traveled by the UAV along its waypoint sequence until the mission is completed.
- **Cooperation Gain**: The ratio of fragments received through intra-cell cooperative exchange versus those received directly from UAV broadcasts. This measures the effectiveness of the cell layer in compensating for link failures.

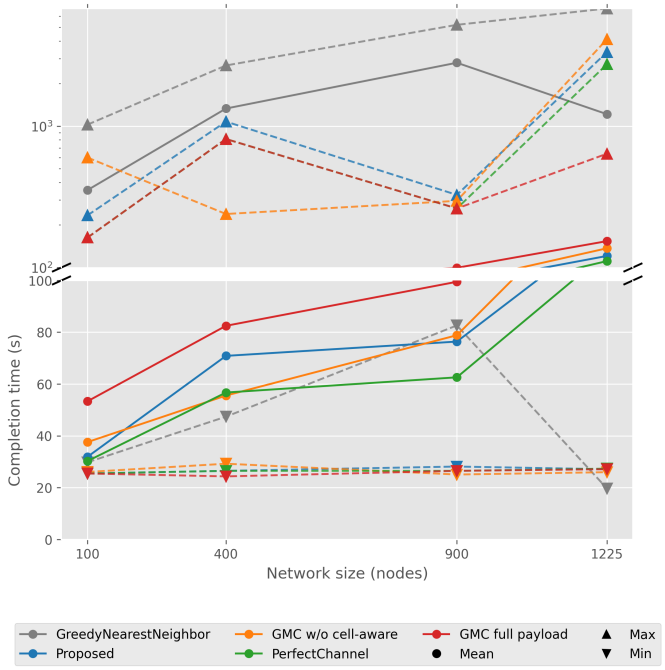

Fig. 3. Detection time (T_{detect}) versus network size for different strategies.

C. Results

We next evaluate detection performance, UAV mobility efficiency, and the role of edge cooperation.

Fig. 3 shows a clear performance gap across strategies. The proposed coordinated fragment dissemination consistently achieves significantly shorter detection times than the nearest-neighbor (NN) baseline, with the advantage becoming more pronounced as network size increases. This indicates that mission efficiency depends more on how quickly useful information is exposed to multiple nodes rather than sequential node visits.

To isolate packet-level impairments, a perfect-channel variant is used as an upper bound. Despite a packet loss rate of approximately 35.3%, the proposed method shows only a small performance gap compared to this ideal case, suggesting that system performance is primarily driven by dissemination and cooperation strategies rather than channel conditions alone.

Fig. 4 compares UAV trajectories under different strategies. Coverage-aware (GMC-based) planning produces shorter and more structured paths than NN. The proposed cell-aware variant further reduces travel distance while maintaining effective coverage, improving overall movement efficiency. This highlights a key trade-off: while detection time may be slightly higher than classical GMC in some cases, the reduction in UAV travel cost leads to a better overall balance between latency and mobility.

The benefit of edge cooperation is illustrated in Fig. 5. The designated detection node can accumulate useful information from neighboring nodes even before direct UAV contact, and

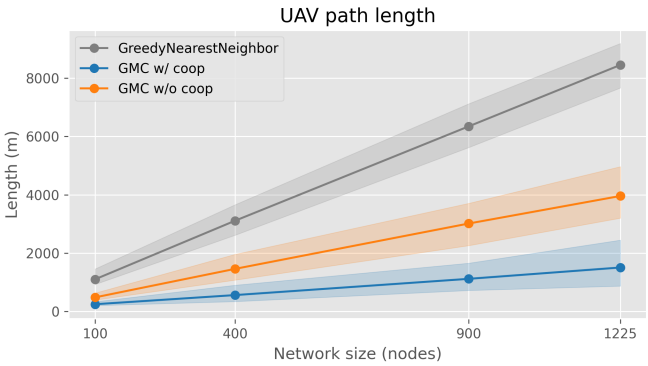

Fig. 4. Comparison of UAV flight trajectories for different strategies.

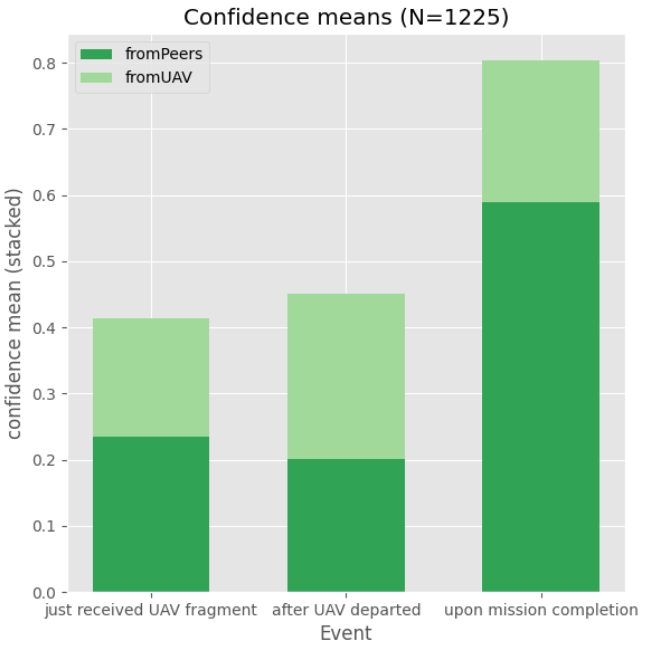

Fig. 5. Accumulation of detection confidence over the mission for $N = 1225$ nodes. The stacked bars show the contribution of fragments received directly from the UAV versus those obtained through peer cooperation.

a substantial portion of the final confidence is contributed by shared fragments. This demonstrates the role of cooperation in amplifying information, particularly in larger networks.

Overall, the results indicate a shift in design strategy: instead of attempting full payload delivery over short and unreliable A2G links, continuous fragment dissemination combined with local cooperation enables steady and efficient information accumulation. These findings support the central premise of this work that jointly coordinating dissemination, trajectory planning, and edge cooperation is key to reducing detection latency under realistic A2G constraints.

VI. Conclusion

This paper studied the problem of minimizing mission-level time-to-detection in UAV-assisted urban IoT sensing systems under realistic packet-level communication constraints. We proposed a coordinated framework that integrates three key mechanisms: payload fragmentation, coverage-aware UAV trajectory planning, and localized edge cooperation. By continuously broadcasting fragments during flight and enabling nodes to exchange missing information, the system accelerates confidence accumulation at the designated detection node without requiring full payload delivery.

The design is built on a realistic A2G communication model incorporating path loss, shadowing, fading, receiver sensitivity, and packet errors, making it suitable for harsh urban environments with short contact windows. Simulation results indicate that the proposed approach reduces detection latency and UAV travel cost compared with conventional node-by-node delivery, while demonstrating the effectiveness of edge cooperation in mitigating unreliable links.

Overall, the results highlight that jointly designing dissemination, mobility, and cooperation is critical for fast target detection in UAV-assisted IoT systems.

Future work includes adaptive trajectory design under dynamic channel conditions, more efficient cooperation control, multi-UAV extensions, and integration with task-oriented fragment generation from real edge-AI models.

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